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Review

Artificial Intelligence in Smart Agriculture Across the Production-to-Postharvest Continuum: Progress, Challenges, and Future Directions

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Abstract

Artificial intelligence is transforming agriculture from a mechanized, labor-intensive sector into a data-driven, perception-enabled, and increasingly autonomous production system. In this review, AI serves as an umbrella term encompassing machine learning, computer vision, and robotic control, among other technologies. We synthesize recent advances across the tillage–sowing–management–harvesting (TSMH) workflow, covering intelligent tillage, precision sowing, field management, and robotic harvesting. The literature shows that AI has significantly improved agricultural perception, prediction, and task-level decision-making. However, large-scale adoption remains constrained by data heterogeneity, limited cross-scene generalization, environmental uncertainty, and insufficient integration across operational stages. Future progress will depend on multimodal data fusion, lightweight and interpretable models, cloud-edge collaboration, and full-chain decision architectures. By framing current research within the TSMH pipeline, this review highlights both technical advances and the critical bottlenecks that must be addressed to move smart agriculture from stage-specific intelligence toward system-level autonomy. Representative studies indicate that AI models can improve soil-property prediction and reduce sowing miss-detection rates to below 3% under controlled or bench-top conditions. However, field deployment may be affected by environmental variability, including illumination changes, dust, vibration, occlusion, and hardware constraints. These limitations highlight the need for robust and edge-compatible architectures.

Keywords: artificial intelligence; digital agriculture; smart agriculture; precision farming; crop monitoring; robotic harvesting; postharvest handling

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1. Introduction

Smart agriculture is increasingly recognized as a strategic pathway for addressing intensifying global pressures on food systems, including population growth, labor shortages, resource scarcity, and climate instability. Its fundamental objective is not merely to increase agricultural output, but to improve production efficiency, product quality, and resource-use sustainability through the deep integration of sensing, computation, and intelligent decision-making technologies. Against this background, the rapid development of artificial intelligence (AI) has markedly expanded the technological boundaries of precision agriculture, enabling more advanced perception, prediction, optimization, and control across the full production cycle, from tillage and sowing to field management and harvesting [1]. As a result, AI is increasingly viewed not simply as an auxiliary tool, but as a core enabling technology for next-generation agricultural modernization and food security [2]. Nevertheless, agricultural production remains one of the most complex application domains for intelligent systems. Crop growth is governed by nonlinear biological processes, while field environments are highly dynamic, spatially heterogeneous, and continuously influenced by weather, soil variability, and operational disturbances. Under such conditions, traditional manual inspection and conventional mechanized operations often struggle to simultaneously achieve accuracy, adaptability, and operational efficiency [3].

The technical demands of agricultural intelligence differ substantially across production stages, yet these stages are tightly coupled in practice. Precision sowing demands rapid seed trait identification, accurate seed-flow monitoring, and dynamic error compensation under high-speed operation [4]. Field management requires coping with fluctuating weather and environmental uncertainty in plant protection, irrigation, and fertilization. Harvesting faces persistent challenges in fruit detection under occlusion, localization in cluttered canopies, and dexterous manipulation [5]. These stage-specific challenges reveal that agricultural AI cannot be adequately understood through isolated task demonstrations alone. Although many recent studies report strong performance in narrowly defined scenarios, most existing systems remain optimized for local objectives and are only weakly connected to upstream and downstream operations. Consequently, the broader problem of building an integrated agricultural intelligence chain—from perception and diagnosis to coordinated actuation and feedback-driven optimization—remains insufficiently addressed.

Despite progress in soil sensing, precision sowing, crop monitoring, water-fertilizer regulation, robotic harvesting, and postharvest handling, the literature remains skewed toward task-specific algorithm design rather than end-to-end system integration. This fragmentation limits model transferability and impedes the development of interoperable smart-farming ecosystems. To address this gap, this review adopts a production-to-postharvest framework that systematically examines AI applications across the tillage–sowing–management–harvesting continuum, while evaluating methodological strengths, recurring constraints, and cross-stage bottlenecks, with particular attention to scalability, robustness, and operational coupling. By reframing existing research within a full-chain perspective, this article aims to provide a clearer conceptual foundation for integrated, data-driven, and increasingly autonomous agricultural systems. The remainder of this paper is organized as follows. Section 2 introduces the literature search strategy and review scope. Section 3 reviews AI applications in tillage and field preparation, including soil property sensing, intelligent tillage machinery, and field path planning. Section 4 focuses on AI-enabled precision sowing and early-stage decision-making, including sowing quality monitoring, seed phenotyping, and sowing-environment analysis. Section 5 discusses AI applications in crop management, with emphasis on plant protection, precision regulation of water, fertilizer, and agrochemical inputs, and crop growth monitoring. Section 6 examines AI technologies in harvesting and postharvest handling, including visual perception for harvesting robots, field harvesting coordination, and quality grading, and traceability. Section 7 synthesizes the major strengths, limitations, and bottlenecks across the full agricultural workflow. Finally, Section 8 concludes the paper and outlines future research directions.

In summary, the present review distinguishes itself from existing surveys in three key respects. While several recent reviews have examined AI applications in specific agricultural domains, such as crop monitoring, robotic harvesting, or precision irrigation, this review adopts a complementary full-chain perspective. First, it adopts a unified TSMH framework that spans the entire production-to-postharvest continuum, enabling systematic identification of cross-stage coupling challenges that are often overlooked in stage-specific surveys. Second, it places equal emphasis on technical advances and operational bottlenecks, with dedicated attention to robustness, scalability, and economic feasibility—dimensions that are frequently underemphasized in algorithm-focused reviews. Third, it synthesizes recent literature from 2021–2025, capturing the rapid evolution of edge-AI, vision transformers, and self-supervised learning in agricultural contexts. By framing current research within a full-chain perspective, this review aims to bridge the gap between isolated technical demonstrations and integrated, field-ready agricultural intelligence.

2. Literature Search Strategy and Review Scope

This review adopts a structured narrative approach to synthesize recent advances in AI applications across the tillage–sowing–management–harvesting continuum in smart agriculture. Relevant studies were retrieved from the Web of Science Core Collection, Scopus, ScienceDirect, and Google Scholar using combinations of the following keywords: "artificial intelligence", "smart agriculture", "precision agriculture", "soil analysis", "tillage", "path planning", "precision sowing", "seed phenotyping", "crop monitoring", "plant protection", "irrigation", "fertilization", "harvesting robot", "yield prediction", and "quality grading". To enhance traceability and minimize topic drift, the retrieved literature was organized using a stage-based screening framework that mapped studies to tillage, sowing, crop management, harvesting, and postharvest handling prior to thematic synthesis.

The literature search primarily covered publications from January 2021 to December 2025, while a limited number of earlier studies were retained when they were methodologically foundational or especially representative of specific research directions. Studies were included if they: (1) focused on AI-related techniques, such as machine learning, deep learning, reinforcement learning, machine vision, hyperspectral imaging, or robotic control; (2) addressed one or more stages of the agricultural production chain, including tillage, sowing, management, harvesting, or postharvest handling; and (3) reported clear agricultural application scenarios, technical workflows, datasets, or performance outcomes. Studies were excluded if they were unrelated to agricultural production processes, lacked substantive AI content, mainly concerned non-agricultural industrial applications, or provided insufficient technical detail for cross-study comparison. Because the reviewed studies differ substantially in crops, sensors, task definitions, and evaluation metrics, the review was designed as a structured qualitative synthesis rather than a meta-analysis. Initial database searches yielded approximately 2450 records (Web of Science Core Collection: 980; Scopus: 870; ScienceDirect: 320; Google Scholar: 280). Duplicate records were identified and removed through manual screening, leaving 1850 unique records. Following title and abstract screening for relevance to the TSMH framework, 1200 full-text articles were assessed for eligibility. Of these, 215 studies met all inclusion criteria and were retained for the final qualitative synthesis. To ensure the quality and relevance of the included literature, studies were further evaluated based on the following quality assessment criterion: (1) clarity of the agricultural application context; (2) technical rigor of the AI methodology description; (3) availability of quantitative performance metrics; and (4) relevance to field-scale deployment considerations. Studies meeting at least three of these four criteria were prioritized for in-depth discussion.

After screening titles, abstracts, and full texts for relevance, the selected studies were coded according to both production stage and technical function, including perception, diagnosis, prediction, decision support, and operational control. Cross-study comparison was then performed using four recurring analytical dimensions: data source, model type, deployment setting, and operational objective. This coding strategy helped distinguish whether reported performance gains mainly reflected improvements in sensing accuracy, decision quality, control stability, or system-level coordination. Because the reviewed literature spans heterogeneous crops, sensors, platforms, and experimental settings, this article follows a structured narrative-review logic rather than a strict meta-analytic design. The emphasis is therefore placed on representative technical advances, cross-stage methodological trends, comparative deployment conditions, and recurring bottlenecks that limit the translation of algorithmic progress into field-scale operational intelligence. Table 1 summarizes the representative tasks, data sources, AI methods, and current bottlenecks across the four main stages of agricultural production.

Table 1. Cross-stage analytical summary of AI applications in the tillage-sowing-management-harvesting workflow.

Stage	Representative Tasks	Typical Data Sources	Common AI Methods	Current Bottlenecks
Tillage and field preparation	Soil sensing, tillage-depth control, obstacle perception, route planning	Spectral data, IoT sensor data, engine/hydraulic signals, UAV or field-environment data	ML/DL regression, CNN-based perception, optimization algorithms, reinforcement learning	Poor transferability across soil types and terrains; weak coupling between sensing, control, and execution
Precision sowing and early-stage decision-making	Seed recognition, sowing-quality monitoring, seed phenotyping, sowing-window prediction	Machine-vision images, metering-device signals, thermal or RGB seed images, weather and soil variables	Object detection, image classification, fuzzy-PID control, hybrid prediction models	Sensitivity to lighting and vibration, limited robustness at high operating speed, weak linkage between phenotype evaluation and field deployment
Crop management	Disease and weed detection, irrigation/fertilization control, growth monitoring, yield prediction	Leaf/canopy images, multispectral or hyperspectral data, soil-moisture sensors, climate records	Segmentation networks, CNNs, transformers, RL-based decision models, multimodal fusion	Heavy annotation costs, data heterogeneity across crops and seasons, insufficient interpretability for management decisions
Harvesting and postharvest handling	Fruit detection, grasp planning, harvesting coordination, grading, traceability	RGB-D images, hyperspectral data, robotic sensor streams, logistics and grading records	Target detection, pose estimation, robotic control, quality prediction, traceability analytics	Occlusion, real-time constraints, and limited harvest-to-grading coordination

3. AI in Tillage and Field Preparation

AI applications in tillage and field preparation have expanded from isolated sensing tasks to more integrated systems that combine perception, machinery control, and route optimization. At this stage, the main objectives are to improve soil information acquisition, enhance operational stability, and support efficient field navigation under complex terrain conditions. Recent studies have demonstrated notable progress in soil property sensing and inversion, adaptive tillage machinery control, and intelligent path planning. At the same time, transferability across soil and terrain conditions, real-time robustness, and effective coupling between sensing, decision-making, and execution remain key constraints. Figure 1 summarizes the representative tasks, typical data sources, common AI methods, and major bottlenecks of AI applications in tillage and field preparation discussed in this section.

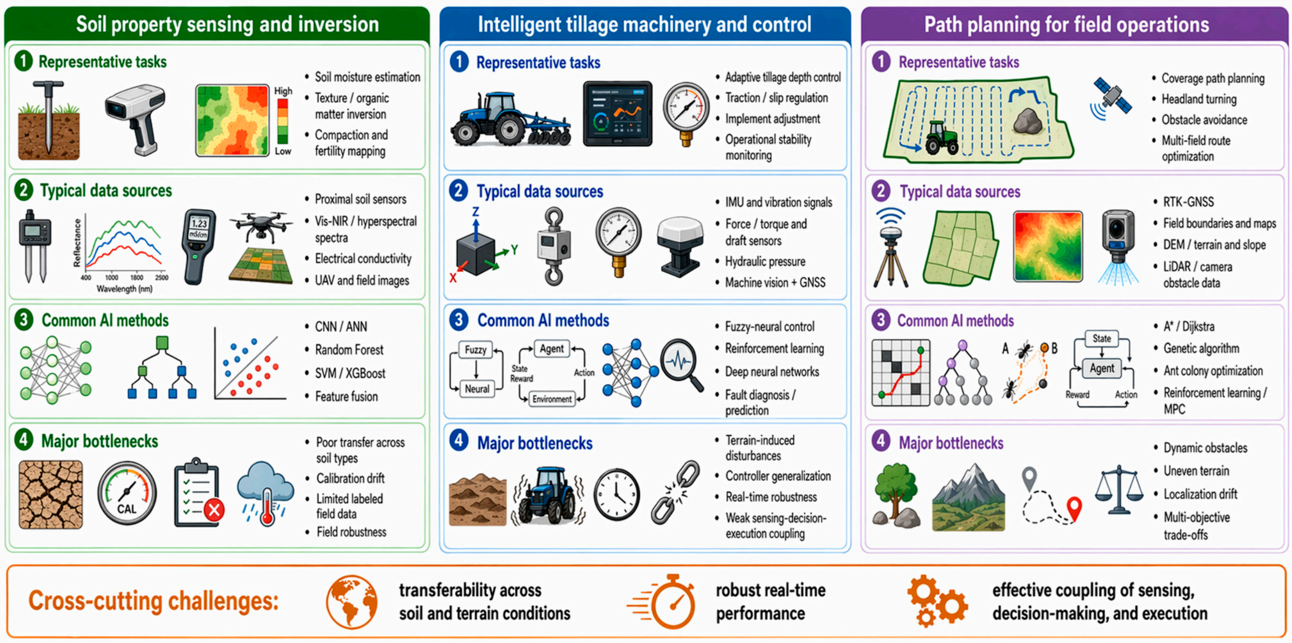

Figure 1. AI applications in tillage and field preparation.

As summarized in Figure 1, current research in this stage can be broadly organized into three major directions: soil property sensing and inversion, intelligent tillage machinery and control, and path planning for field operations.

3.1. Soil Property Sensing and Inversion

The growing demand for high-quality agricultural data is reshaping how soil properties are acquired and interpreted. Rapid, non-destructive sensing methods that combine spectroscopy with AI are gradually supplementing, and in some cases replacing, conventional laboratory-based chemical analysis. Compared with traditional linear-regression methods, AI-driven models are better able to capture the nonlinear relationships between soil properties and environmental covariates, thereby improving prediction accuracy and practical applicability in soil science and precision agriculture.

To further exploit deep spectral features, Datta et al. compared multiple machine-learning and deep-learning models for predicting soil properties from visible-spectrum data and found that the optimal model differed across target variables. For some indicators, random forest and support vector regression achieved higher predictive accuracy, whereas 1 D-CNN showed more stable error behavior without consistently outperforming the other models. The study also evaluated the influence of feature-transformation methods such as PCA and EMD on model performance, providing useful guidance for model selection and feature construction in low-cost rapid soil testing [6]. However, when laboratory-calibrated spectral inversion models are transferred to in situ field monitoring, changes in moisture and other environmental conditions can substantially alter reflectance spectra and absorption characteristics, thereby weakening model robustness and transferability. To address this issue, Huang et al. introduced the concept of a spectrally effective moisture threshold into an exponential spectral reflectance model and proposed an improved method for simulating equivalent spectral reflectance under different moisture conditions. This work provides methodological support for estimating soil salinity and related properties under more realistic field conditions [7].

3.2. Intelligent Tillage Machinery and Control

The intelligence level of tillage and land-preparation operations is closely linked to seedbed quality and subsequent crop growth. Current technologies are evolving from isolated mechanical control toward adaptive decision-making systems supported by multi-source sensing. In their review of smart sensors and data processing, Soussi et al. noted that next-generation sensor systems are no longer confined to measuring conventional physical variables; rather, they are increasingly developing into intelligent sensing terminals with edge-computing capability. This shift is highly relevant to tillage machinery because it enables high-precision real-time data acquisition and local processing, which can directly support operations such as tillage-depth calibration on uneven terrain and dynamic monitoring of soil fragmentation [1].

Building on these sensing capabilities, real-time interaction between machinery and the operating environment has become a central issue in intelligent equipment management. Yin et al. developed an edge-computing-based Internet of Things (IoT) monitoring system for agricultural machinery. At the vehicle level, sensing devices together with attitude and operational-parameter algorithms were used to collect and process subsoiling parameters in real time, while a cloud-based interaction mechanism ensured stable data transmission between field equipment and the cloud platform. The cloud side then supported operation statistics, quality evaluation, and efficiency analysis, thereby forming a closed-loop framework of perception, transmission, and decision-making for agricultural machinery management [8].

To address load fluctuations during tillage, Han et al. established a tillage-depth control system based on engine load characteristics. By estimating engine load from throttle opening, engine speed, and load-characteristic models, the system can automatically raise the plow when soil resistance increases and engine speed deviates from the target range, thereby reducing tillage depth and traction resistance while stabilizing engine performance and improving fuel economy [9]. For hydraulic hitch adjustment, Luo et al. addressed the challenge of simultaneously maintaining constant tillage depth and desirable traction performance by constructing a tractor-implement dynamic model and using a genetic algorithm to compute the optimal hydraulic-cylinder pressure online. Their results showed that the system could maintain tillage depth while improving traction efficiency, reducing wheel slip, and lowering overall energy loss [10]. In a related study, Luo et al. proposed an energy-saving drive-control strategy for electric tractors based on terrain-parameter identification. By identifying terrain parameters for the left and right drive wheels in real time and calculating the optimal slip ratio, a feedforward PID torque controller maintained the slip ratio within an optimal range, thereby reducing energy losses caused by excessive slip and improving speed stability during operation [11].

With the increasing application of autonomous navigation and machine vision, Syed et al. further developed a real-time convolutional neural network (CNN)-based obstacle-classification method for autonomous orchard robots. This method supports obstacle avoidance and motion decision-making in complex agricultural environments and illustrates how deep-learning-based visual perception is helping agricultural robots move from assisted automation toward higher levels of autonomy [12].

3.3. Path Planning for Field Operations

Path planning under complex terrain is critical for reducing energy consumption, minimizing soil compaction, and improving operational efficiency. Current research is moving beyond conventional geometric coverage methods toward intelligent decision systems with multi-objective optimization capability. He et al. proposed an automatic boundary-line detection method for complex paddy-field scenes based on deep-learning segmentation and geometric fitting. Because the input imagery included noise such as wheel ruts, shadows, weeds, and uneven illumination, the study demonstrated the feasibility of identifying field and operation boundaries in unstructured farmland, thereby providing essential prior information for subsequent path generation and operation planning [13].

In route planning for multiple fields involving irregular plots and heterogeneous machinery, Utamima et al. treated the minimization of nonworking distance as the core objective and incorporated constraints such as machinery maneuvers into the optimization framework. Their results showed that reducing machinery transfer distance and ineffective in-field travel is important not only for lowering operational costs, but also for decreasing fuel-related carbon emissions [14].

At the regional scale, Donat et al. established a technical workflow including cultivation-direction identification, path design, and path evaluation. By comparing current and optimized operations across large numbers of fields, they quantified the potential reduction in headland area and turning frequency, thereby revealing the macro-level benefits of reducing ineffective operations through workflow optimization [15].

To meet the need for complete coverage in irregular plots, Han et al. developed an improved genetic algorithm incorporating a Markov-chain state-transition mechanism and applied it to complete-coverage path planning. Simulation experiments under different operational settings showed that the method significantly reduced path length, number of turns, and repeated-operation rate while improving coverage performance, indicating that evolutionary and heuristic search strategies can adapt flexibly to the geometric complexity of farmland [16].

In hilly terrain involving cooperative operation by multiple machines, Liu et al. constructed a terrain-feature model based on a digital elevation model to achieve regional complete-coverage path planning. They further combined Dijkstra's algorithm with an improved Harris Hawk optimization algorithm to build a multi-objective collaborative scheduling model aimed at reducing fuel consumption, scheduling cost, and operation time, thereby directly addressing the practical challenges of complete coverage and resource allocation in hilly agricultural areas [17].

Beyond static global optimization, path-planning systems are increasingly expected to perceive environmental states in real time and dynamically replan routes in response to obstacles and unexpected disturbances. Gautron et al. pointed out that reinforcement learning can learn optimal crop-management strategies through interaction with simulated environments [18]. In a related study, Li et al. investigated multi-area coverage path planning based on an improved double deep Q-network and designed a multi-objective reinforcement-learning reward function with penalties and incentives. Their results showed that the method reduced redundant paths while improving overall coverage efficiency in complex operations [19]. Taken together, these studies indicate that path-planning research is shifting from offline route optimization toward perception-aware and adaptive decision-making. Nevertheless, robust deployment still depends on accurate terrain representation, stable online perception, and computationally efficient replanning under field uncertainty.

Overall, recent progress in tillage-stage AI indicates a transition from isolated sensing or route-design tools toward integrated perception-control systems. The remaining challenge is no longer simply improving local sensing accuracy or routing efficiency, but linking these capabilities to downstream sowing decisions through interoperable field maps, machinery-state data, and seedbed-quality indicators.

These stage-level advances, however, remain largely disconnected from downstream sowing decisions—a gap that the cross-stage synthesis in Section 7 addresses in more detail.

4. AI in Precision Sowing and Early-Stage Decision-Making

AI-enabled precision sowing is increasingly evolving from simple monitoring toward a closed-loop framework that integrates seed perception, equipment regulation, and environment-aware decision-making. This stage is particularly important because sowing quality directly affects crop emergence uniformity, subsequent field management, and final yield formation. Recent research has focused on sowing-quality monitoring, seed phenotyping, and sowing-window decision support, showing clear progress in both sensing accuracy and control capability. However, robustness under vibration, variable illumination, dust interference, and heterogeneous seed traits remains a major challenge for large-scale field deployment. Figure 2 provides a conceptual summary of the main tasks, data inputs, AI methods, and operational constraints involved in precision sowing and early-stage decision-making.

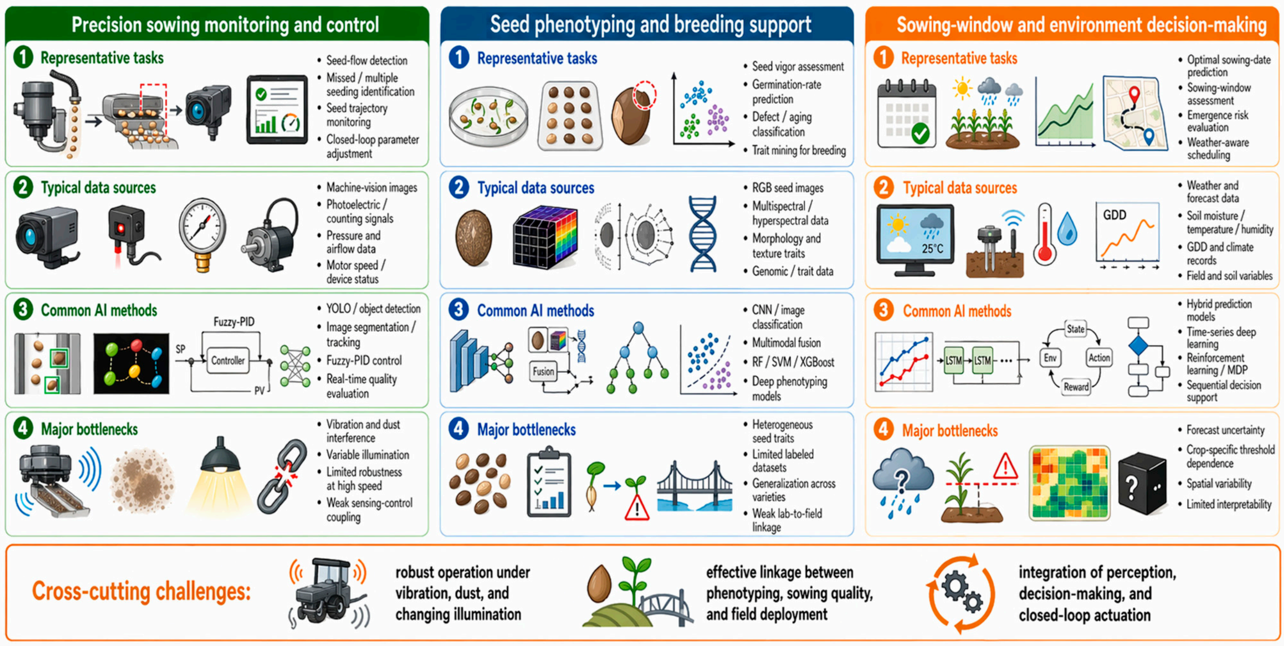

Figure 2. AI applications in precision sowing and early-stage decision-making.

As shown in Figure 2, AI applications in this stage mainly include precision sowing monitoring and control, seed phenotyping and breeding support, and sowing-window and environment-related decision-making.

4.1. Precision Sowing Monitoring and Closed-Loop Control

The core objective of precision sowing is to reduce the mechanical randomness of seed-metering devices under high-speed operating conditions and to ensure accurate seed placement and uniform spatial distribution. Traditional monitoring methods rely mainly on photoelectric sensors; however, under dusty field conditions or when handling very small seeds, signal attenuation and limited resolution can substantially increase false detections. To address this issue, Zhang et al. developed an online monitoring architecture for high-speed flow detection of small vegetable seeds that integrates image acquisition, feature segmentation, and missed-seeding identification [20].

In AI-based visual recognition, precision is a standard evaluation metric for characterizing false-detection risk and model reliability. In their review on deep learning and computer vision for plant disease detection, Upadhyay et al. identify *Precision* as a key indicator of visual recognition system reliability under complex agricultural backgrounds [21]. The calculation formula is given in Equation (1).

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

In the context of high-speed precision sowing monitoring, *Precision* represents the proportion of correctly identified seeds among all positive detections. Values of precision below certain thresholds in field conditions typically indicate the need for additional filtering algorithms due to false positives caused by dust, soil clods, or crop residue. For seed-detection tasks, *Precision* represents the proportion of true seed particles among all samples identified as positive; True Positives (*TP*) denotes the number of correctly recognized seed particles, and False Positives (*FP)* denotes impurities or background noise mistakenly classified as seeds.

For the visual inspection of pneumatic seed metering, Xing et al. developed a real-time sowing-accuracy detection system consisting of a negative-pressure fan, driving motor, pneumatic seed-metering device, pressure sensor, industrial camera, and motor controller. An improved YOLOv5 n model was embedded into the system to achieve real-time recognition of small seeds under low-illumination conditions inside the device. Their experiments verified detection accuracy under different suction-disk rotational speeds and fixed adsorption negative pressures, thereby providing direct evidence for process monitoring and parameter adjustment [22]. On a test bench for air-suction seed-metering devices, Zhang et al. further developed an integrated monitoring-and-control scheme in which real-time counting was achieved through photoelectric sensing, while image-recognition technology was used to dynamically monitor sowing states and seed trajectories, including missed and repeated seeding. The experimental results provided quantitative indicators for the accuracy of detecting missed and repeated seeding, confirming the effectiveness of visual monitoring for evaluating sowing quality [23]. As visual perception accuracy continues to improve, research emphasis is gradually shifting from state recognition to closed-loop control, with the goal of developing adaptive systems capable of adjusting seed-metering parameters in real time according to sowing quality. For negative-pressure stability control, He et al. proposed an adaptive negative-pressure control system for electric-driven sowing blowers based on a DILPSO-fuzzy PID framework, thereby offering a control-layer solution for translating visual or sensor-derived information into negative-pressure stabilization and closed-loop parameter optimization [24].

4.2. Seed Phenotyping and Intelligent Breeding Support

Intelligent breeding provides an important pathway for seed-quality grading by enabling rapid and non-destructive evaluation based on seed vigor and phenotypic traits. Nehoshtan et al. advanced seed-germination prediction by reconstructing traditional vigor-testing workflows with deep learning. Their results showed that convolutional neural networks can learn discriminative features from RGB image data and accurately capture morphological changes during early germination stages, thereby enabling high-accuracy prediction of germination rates before actual germination occurs and providing a useful computational tool for shortening breeding cycles and screening high-vigor germplasm [25]. Traditional manual breeding inspection is time-consuming, labor-intensive, and subjective, making it unsuitable for the high-throughput screening required by modern seed industries. In research on rice seed vigor testing, Qiao et al. systematically demonstrated the value of computer vision for quantitative micro-phenotyping. By integrating multispectral imaging with machine-learning algorithms, their study simultaneously detected internal biochemical components and external morphological defects of rice seeds, showing that multimodal visual-data fusion can significantly improve the discrimination of seed aging and stress resistance and thus provide a technical foundation for refined seed grading [26]. With the continuous development of phenomics, research attention has expanded from single-task vigor detection to multidimensional phenotypic analysis covering morphology, texture, and genetic traits, with the broader objective of constructing accurate genotype-phenotype mappings. In a comprehensive study on the use of AI and machine learning for mining superior crop genes, Kassem pointed out that traditional statistical methods have clear limitations when analyzing agronomic traits controlled by complex polygenic systems. By reviewing the application of machine-learning algorithms for identifying quantitative trait loci associated with seed-quality traits, the study showed that AI models can overcome important bottlenecks of conventional genetic analysis in multidimensional data integration through strong nonlinear pattern-recognition capability and the capacity to process large-scale genomic data [27].

4.3. Sowing-Window and Environment Decision-Making

Determining the optimal sowing time requires transforming multidimensional environmental variables into a quantifiable model of the operational window. In their review of Growing Degree Days (*GDD*) models from flowering to harvest, Fotouo Makouate and Zude-Sasse pointed out that *GDD* is one of the key phenotypic indicators used to describe crop responses to temperature. In agricultural decision-making, *GDD* represents the cumulative heat units available for crop development and is commonly used to estimate phenological progress, field-work windows, and harvest timing. The calculation formula is given in Equation (2) [28].

$$GDD = \sum \frac{(T_{max} + T_{min})}{2} - T_{base} \tag{2}$$

In agricultural practice, T_{base} is crop-specific (e.g., 10 °C for maize). The *GDD* equation is applied to predict optimal sowing windows; a cumulative *GDD* value within the range of 100–150 °C·d following sowing is considered a critical threshold for achieving uniform crop emergence. In this framework, T_{max} and T_{min} represent the daily maximum and minimum temperatures, respectively, and T_{base} denotes the lower threshold temperature required for crop development.

In research on deep-learning frameworks for precision-agriculture optimization, Logeshwaran et al. proposed an agro-deep learning framework capable of processing environmental variables such as soil moisture, temperature, and humidity. By integrating these variables into a unified prediction framework, the model aims to better characterize crop growth behavior and improve decision-making in precision agriculture [29]. Because sowing-environment decisions typically involve long cycles, cumulative environmental effects, and stage-specific physiological thresholds, sequential decision methods are increasingly relevant. Although Chen et al. focused on irrigation rather than sowing directly, their reinforcement-learning formulation under uncertain weather provides a useful methodological reference for agricultural time-window optimization. By modeling the decision process as a Markov decision process and incorporating short-term weather forecasts, they demonstrated how dynamic policies can balance resource efficiency with yield stability [30]. Jabed and Murad, in their review of crop-yield prediction, further emphasized that machine-learning and deep-learning methods are especially valuable for handling high-dimensional agricultural data and supporting management decisions under climate variability [31]. In summary, sowing-window decision-making is evolving from experience-based calendar judgment toward data-driven sequential optimization. Its main remaining bottleneck lies in integrating short-term environmental forecasts with crop-specific physiological thresholds in a way that is both robust and operationally interpretable.

Overall, AI applications at the sowing stage are increasingly moving toward a closed-loop paradigm that connects seed perception, equipment regulation, and environmental decision-making. However, robustness under high-speed field vibration, changing illumination, and heterogeneous seed traits remain a major constraint on large-scale deployment. More importantly, the quality of sowing-stage decisions directly shapes subsequent crop-emergence uniformity, field-management scheduling, and yield-formation potential, making this stage a critical bridge between tillage preparation and in-season management.

The closed-loop framework described here represents a critical step toward integrated precision agriculture, yet its full potential depends on standardized data interfaces with upstream and downstream stages, as discussed in Section 7.

5. AI in Crop Management

In crop management, AI is increasingly used to support diagnosis, prediction, and prescriptive intervention across plant protection, input regulation, and growth monitoring. Compared with earlier production stages, management tasks usually involve more heterogeneous data sources, stronger temporal dynamics, and closer coupling between perception and action. Current studies have demonstrated substantial progress in disease and weed detection, irrigation and fertilization support, and crop growth assessment through the integration of multi-source sensing, deep learning, and decision models. Nevertheless, annotation cost, cross-season and cross-crop heterogeneity, limited interpretability, and deployment complexity remain persistent limitations. Figure 3 summarizes the representative management tasks, sensing inputs, methodological categories, and key operational outputs covered in this section.

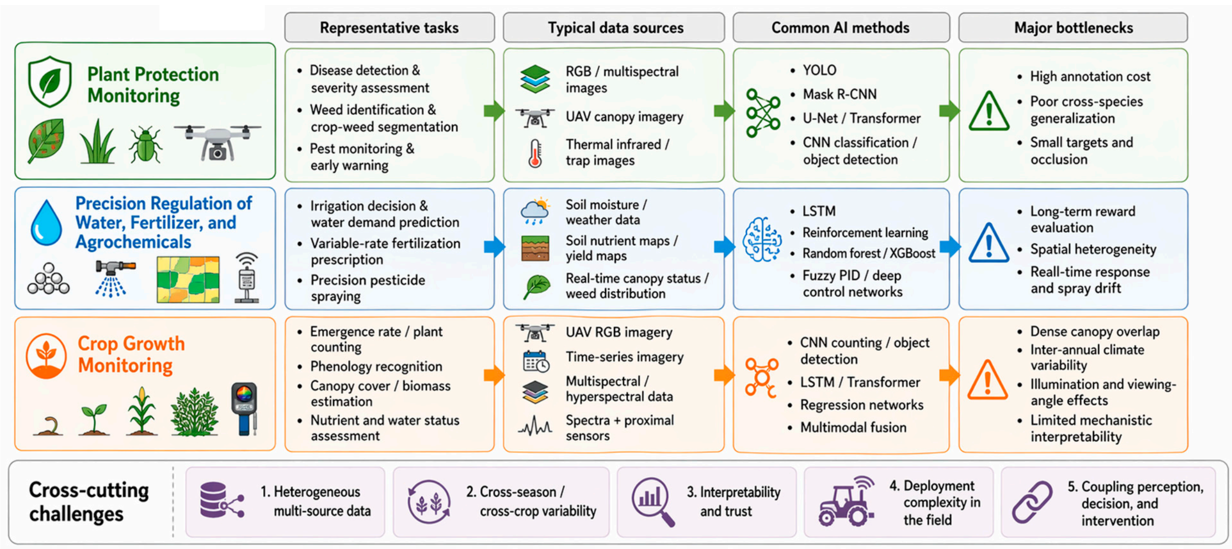

Figure 3. AI applications in crop management.

As illustrated in Figure 3, AI-driven crop management can be organized around three major themes: plant protection monitoring, precision regulation of water, fertilizer, and agrochemical inputs, and crop growth monitoring.

5.1. Plant Protection Monitoring

Plant protection monitoring is an important means of ensuring healthy crop growth and controlling the use of fertilizers and pesticides. It shifts inspection from traditional experience-based manual scouting to intelligent management through automatic recognition systems based on multi-source sensing and deep learning. In research on optimizing deep learning for agricultural visual recognition tasks, target discrimination in complex field environments remains difficult. To address this problem, Hu et al. proposed an improved deep residual network, SkipResNet, for efficient crop and weed recognition [32].

In deep-network applications, processing multidimensional features is crucial. In addition, in the field of health monitoring for grain crops, Syed et al. focused on leaf disease detection and classification and showed that introducing efficient feature-dimensionality reduction can significantly optimize the processing of plant leaf images and improve overall disease-classification performance [33]. To meet the need for high-accuracy phenotypic recognition under limited labeled data, Zhan et al. proposed the use of semantic-segmentation models for pixel-level classification of crops and segmentation and used intersection over union (*IoU*) to evaluate segmentation accuracy [34]. The calculation formula is given in Equation (3).

$$IoU = \frac{A \cap B}{A \cup B} \tag{3}$$

IoU is widely used to evaluate segmentation reliability in weed and crop recognition tasks, with higher *IoU* values indicating better spatial alignment between predicted and ground-truth regions. In precision agriculture applications, appropriate *IoU* thresholds are typically determined based on task-specific requirements and operational constraints. *IoU* measures the overlap ratio between prediction and reference and is therefore widely used to assess segmentation reliability in crop-protection and field-monitoring tasks [35].

In research on deep-learning architectures for precision agriculture, Muthusamy and Ramu focused on how to achieve accurate identification of crop diseases and nutritional problems by optimizing deep neural network models. Specifically, after introducing dynamic attention mechanisms and efficient feature-network modules, the deep-learning system could not only perform efficient phenotypic analysis of crop diseases, but also conduct fine-grained severity classification and high-accuracy discrimination of micronutrient deficiencies and pathological diseases [36,37]. For specific agricultural production scenarios such as hydroponics, deep-learning-based dynamic models have also been effectively applied to real-time detection of plant nutrient status [38]. In a comprehensive study covering multiple tasks including pest and disease diagnosis, irrigation scheduling, and quality evaluation, Ajith et al. thoroughly discussed the use of machine-learning and deep-learning algorithms in several core tasks of precision agriculture, clearly indicating that these advanced models can effectively support crop-yield prediction, precise pest and disease diagnosis, soil-fertility mapping, refined irrigation scheduling, and food-quality assessment [39].

To improve the systematic level of plant protection monitoring, Partel et al. proposed a smart orchard spraying system that integrates multi-sensor fusion with AI-based decision-making. The system can obtain information such as canopy volume and canopy density in real time and then automatically adjust nozzle switching and spray-volume allocation, thereby reducing spray drift and waste while improving the uniformity of canopy coverage [40]. To address the challenges of complex canopy structures and limited inter-row space in orchards, intelligent processing platforms have been built around a closed-loop pipeline of perception, recognition, decision, and execution. In this framework, sensor data on the operational target are transformed by AI models into actionable spraying features, and then into executable parameters via controllers, enabling a transition from rule-based plant protection to coordinated optimization driven jointly by data and models [41,42].

5.2. Precision Regulation of Water, Fertilizer, and Agrochemical Inputs

By introducing multi-source sensing and data-fusion technologies, the perceptual range of agricultural management has been continuously expanded. This change has widened the state space and increased the coupling complexity of coordinated input regulation, thereby imposing stricter requirements on comprehensive algorithmic modeling capability, feature-fusion performance, and decision efficiency [1,2]. The development of precision regulation has shown a clear iterative trajectory: it began with simple threshold-based logical control and has gradually shifted toward intelligent decision-making that combines multi-source environmental perception, crop-state recognition, and adaptive actuation. In practical field deployment, however, water, fertilizer, and agrochemical decisions are often coupled rather than isolated, because irrigation status, nutrient demand, canopy condition, and pest pressure jointly shape management timing and dosage.

Recent studies on deep-learning architectures in precision agriculture have highlighted the importance of hierarchical feature extraction and convolutional networks for processing crop-health images. By integrating optimized deep learning models with multimodal feature fusion techniques, these systems can effectively identify abnormal features caused by pathological factors or environmental stress, while preserving model interpretability. They have also demonstrated strong performance in classifying physiological nutrient deficiencies in plants with relatively high accuracy and discriminative capability [43–45]. However, water and fertilizer management typically involves sequential decision-making and continuous state updates, rather than one-time classification. From a management perspective, the true value of AI lies in linking diagnosis to action—translating sensed crop conditions into timely irrigation, fertilization, or protection prescriptions under dynamic field constraints. To address this sequential decision problem, Ding et al. formalized the decision process as a Markov decision process (MDP) within an incremental reinforcement learning framework and introduced a dual-adaptive epsilon-greedy exploration strategy. They characterized long-term return optimality using the Bellman optimality equation for value functions, offering a principled approach to adaptive management under uncertainty [46].

In recent years, weed recognition has evolved from traditional handcrafted-feature or shallow-classification methods to end-to-end detection and segmentation centered on deep neural networks, and it has become linked with UAV remote sensing to form an operational chain of recognition-prescription-variable-rate spraying. Kebede et al. used UAV imagery and deep learning to detect weeds in teff fields, thereby providing target locations and spatial ranges for precision herbicide application; this workflow can be directly connected to site-specific spraying strategies or inter-row differential spraying [47]. Furthermore, the WeedSwin model proposed by Islam et al. improved weed identification under complex backgrounds and dense plant distributions, indicating that agrochemical-input regulation increasingly depends on accurate pixel-level perception before actuation [48]. For saffron fields, Makarian et al. enhanced the robustness of weed recognition against illumination changes and occlusion by improving EfficientNetB0, thereby improving the model's generalization ability under field conditions [49].

5.3. Crop Growth Monitoring

Regarding recent advances in crop growth monitoring, Ma et al. proposed a method that combines deep multitask learning with a mechanistic model to predict key crop physiological indicators from UAV hyperspectral images, effectively improving the inversion accuracy of important biochemical parameters such as wheat leaf nitrogen content. This provides refined data support for nutritional diagnosis and real-time topdressing in the early stages of crop growth [50]. Kaiser et al. highlighted the importance of multidimensional growth indicators in output-quality evaluation and confirmed the key bridging role of real-time monitoring frameworks in coordinating crop metabolic activities with dynamic environmental interactions [51]. From the technological essence and core objectives of crop growth monitoring, the main goal is to dynamically monitor crop health and preliminarily estimate final yield. The realization of this goal requires continuous tracking of leaf area index (LAI), biomass, and leaf nutrient status, thereby constructing digital models that reflect crop growth and development patterns. Recent studies have demonstrated the value of incorporating physiological indicators—such as canopy water content and chlorophyll content—into remote sensing-based crop monitoring workflows. These indicators help characterize differences in crop growth status and reduce growth-related estimation errors [52].

In summary, AI applications in crop management have progressed from early diagnostic tools to integrated decision-support systems that combine multi-source sensing, predictive modeling, and prescriptive actuation. The convergence of UAV-based imaging, edge-deployed deep learning models, and variable-rate application technologies has enabled increasingly precise and automated management of water, nutrients, and crop protection inputs. Nevertheless, the full potential of these technologies remains constrained by data heterogeneity, limited cross-dataset generalization, and the practical challenges of deploying computationally intensive models in resource-limited field environments—themes that are explored further in the cross-stage synthesis that follows.

6. AI in Harvesting and Postharvest Handling

AI applications in harvesting and postharvest handling extend agricultural intelligence from field execution to downstream grading, traceability, and logistics-related decision support. At this stage, the focus shifts from in-season diagnosis and regulation toward target perception, robotic manipulation, harvest coordination, quality assessment, and information continuity after harvest. Recent research has shown notable advances in harvesting robots, field route optimization, and postharvest grading systems, indicating the growing potential of AI across the harvest-to-market chain. However, major barriers remain in environmental robustness, commercial deployment, and interoperability between field-side sensing systems and downstream postharvest management platforms. Figure 4 presents a comparative summary of the main application directions, data sources, AI methods, and current bottlenecks in harvesting and postharvest handling.

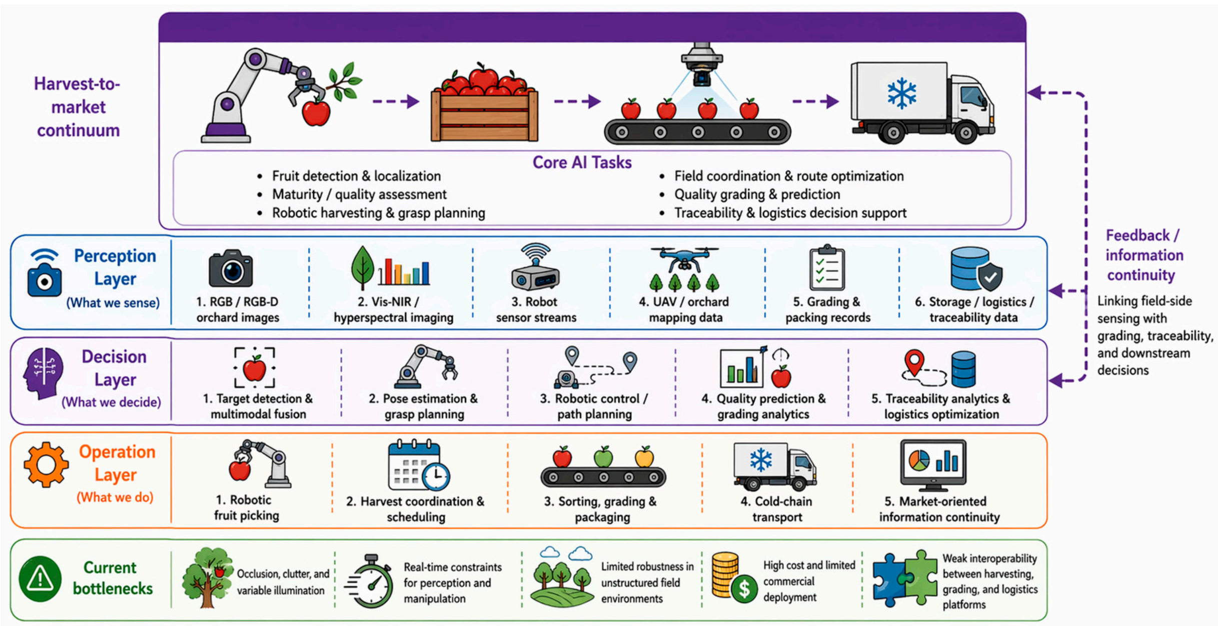

Figure 4. AI applications in harvesting and postharvest handling.

As shown in Figure 4, this stage includes three closely related directions: vision and control of harvesting robots, field harvesting coordination and route optimization, and postharvest quality grading and traceability.

6.1. Vision and Control of Harvesting Robots

Intelligent harvesting robots are key technologies for reducing labor costs and improving operational efficiency in smart orchards. Their central technical challenge lies in achieving high-precision target perception and dexterous robotic manipulation in complex, unstructured environments. To address maturity discrimination, which directly affects the harvest window, Liu et al. showed that multimodal fusion of image and sensing information with deep neural networks can accurately identify the physiological development stage of fruits, thereby reducing the risk of harvesting immature products and improving the consistency of harvest quality [53].

Shao et al. used Vis-NIR hyperspectral imaging for non-destructive tomato-quality assessment and established a Comprehensive Quality Index to quantify integrated changes in multiple internal and external quality indicators during ripening. Their workflow first analyzed variation trends and correlations among quality indicators, then derived the comprehensive index through factor analysis, and finally completed CQI prediction and spatial visualization through feature-wavelength selection and regression modeling. This process demonstrates how visual and spectral information can be transformed into quantitative indicators for maturity evaluation and harvest-time determination [54].

In their review of global progress and future challenges in harvesting robots, Tan et al. noted that although visual recognition accuracy has improved substantially, real-time path planning and dynamic obstacle avoidance remain major bottlenecks limiting commercialization [55]. In related work on harvesting-robot grasping control, Wang et al. proposed a geometry-aware grasping estimation method for apple-orchard robots. By accurately identifying the three-dimensional geometric features of fruits, the method significantly improved the manipulator's grasp success rate in confined spaces [56]. These studies suggest that harvesting systems have progressed more rapidly in target perception than in fully reliable field execution. The key challenge is no longer merely whether fruits can be detected, but whether perception, motion planning, and compliant manipulation can be coordinated robustly under occlusion, deformation, and dynamic disturbance.

Despite these technical advances, the widespread adoption of robotic harvesting systems remains constrained by economic and practical factors. The capital cost of fully autonomous harvesting platforms remains prohibitive for many small to medium-sized orchard operations, with return-on-investment timelines often exceeding five years. Reliability challenges also persist, particularly in unstructured orchard environments where variable lighting, occlusion, and fruit clustering continue to degrade grasping success rates. Furthermore, the mechanical delicacy required for soft-fruit harvesting has yet to be reliably achieved at commercial speeds. These deployment realities underscore the gap between research-grade demonstrations and commercially viable, field-ready harvesting solutions.

6.2. Field Harvesting Coordination and Route Optimization

Efficient harvesting coordination and route optimization rely on spatially explicit crop data and machine-readable field maps. In scenarios requiring fine-grained crop identification, Wu et al. developed a semantic segmentation model based on multi-source remote-sensing images to accurately delineate the spatial distribution of winter wheat at the tillering stage. This approach overcomes the limitations of single-source data and significantly enhances the precision of harvest planning, machinery dispatch, and field-level digital management. This large-scale extraction method, which combines multi-source remote sensing with semantic segmentation, overcomes key limitations of single data sources and substantially improves the accuracy of crop-boundary extraction, thereby providing a data basis for harvest planning, machinery dispatch, and spatial digital management [57]. In the field of large-scale monitoring of plant and crop distributions, deep-learning models have played an important role in multi-temporal mapping and multi-source feature extraction. A recent systematic review has demonstrated the strong potential of deep-learning models to process remote-sensing data for agricultural crop monitoring. For example, Joshi et al. systematically reviewed the integration of remote-sensing data and deep-learning techniques for crop mapping and yield prediction, showing that deep-learning models can represent complex features and nonlinear relationships that are essential for crop mapping and preharvest yield prediction [58].

For orchard management and harvesting, the sequence of tree localization, canopy extraction, and tree counting provides the basis for constructing digital orchard maps and planning field operations. Kelly et al. compared object-based image analysis with several CNN-based object-detection methods for orchard-tree mapping across an entire growing season. Their results showed that deep detection methods perform better under complex backgrounds, occlusion, and canopy-shape variation, whereas object-based approaches still retain some advantages in interpretability and deployment convenience when rules are clear and object boundaries are distinct [59]. These findings are directly relevant to row-level route planning, machine scheduling, and harvest-capacity estimation, and they suggest that future system integration should carefully balance accuracy, efficiency, and interpretability.

6.3. Postharvest Quality Grading and Traceability

Postharvest quality grading is a key link between agricultural production and market realization because it determines how harvested products are classified, priced, and routed during storage and distribution. In recent years, AI-enabled grading systems have shifted from subjective manual inspection toward non-destructive, data-driven assessment based on image analysis, spectral sensing, and multimodal fusion. Recent reviews have specifically highlighted how AI-powered non-destructive technologies, including RGB imaging and Vis-NIR spectroscopy, are increasingly applied to automate postharvest quality grading and disease identification in horticultural crops [60]. In tomato-quality assessment, for example, Shao et al. established a Comprehensive Quality Index based on Vis-NIR hyperspectral imaging and regression modeling, showing how multiple internal and external indicators can be integrated into an objective grading framework [54].

Beyond grading accuracy itself, traceability is becoming increasingly important in postharvest systems. Once quality information has been digitized, AI models can support standardized record generation, batch classification, and quality-state tracking across storage, transport, and marketing stages. In this sense, postharvest AI is valuable not only for identifying differences in appearance or maturity, but also for linking field-side quality perception with downstream traceability management. Future research should therefore place greater emphasis on interoperable quality datasets, multimodal grading models, and digital traceability pipelines capable of connecting harvesting, grading, and market circulation into a coherent information chain. Compared with upstream field applications, postharvest AI currently benefits from more controllable environments and clearer quality labels; however, its broader value will depend on whether grading outputs can be seamlessly connected with logistics, traceability, and market-side decision systems.

Beyond grading, AI is increasingly applied to postharvest logistics and traceability. Recent reviews have highlighted the role of AI-enabled technologies—including machine vision, electronic noses, and near-infrared spectroscopy—in fruit and vegetable quality inspection, cold chain environmental monitoring, and shelf-life prediction [61]. These approaches support dynamic shelf-life estimation and cold chain optimization by integrating real-time sensor data with predictive models. Computer vision systems deployed at packing facilities further enable automated defect sorting and foreign object detection at high throughput. Blockchain-based traceability platforms have also been systematically reviewed for agricultural products, with studies highlighting their potential to improve transparency and data integrity in supply chain management [62]. When coupled with AI-driven quality prediction at harvest, these platforms can help downstream stakeholders anticipate product condition upon arrival and adjust inventory rotation accordingly. However, the integration of field-side quality data with postharvest management systems remains nascent, largely due to interoperability barriers between on-farm sensors and commercial logistics software. Future research should prioritize the development of standardized data exchange protocols and integrated platforms that bridge the harvest-to-market information gap.

The challenges identified at this stage—including environmental robustness, cost constraints, and cross-platform interoperability—are recurring themes that the cross-stage synthesis in Section 7 examines from a broader perspective.

7. Cross-Stage Synthesis: Strengths, Limitations, and Bottlenecks

Following the analytical framework outlined in Section 2, this synthesis is organized around four cross-cutting dimensions: data source, model type, deployment setting, and operational objective. Table 2 provides a consolidated cross-stage comparison across these dimensions, revealing several recurrent patterns. First, RGB and multispectral imagery dominate visual perception tasks across tillage, sowing, management, and harvesting, whereas hyperspectral and IoT sensor data are more prevalent in soil and postharvest quality assessment. Second, CNN-based architectures remain the workhorse for perception tasks, but transformer-based models are gaining traction in segmentation and multimodal fusion. Third, deployment settings shift from cloud-based training to edge inference as the application moves from offline planning (e.g., path optimization) to real-time control (e.g., seed metering compensation). Fourth, operational objectives evolve from predictive diagnostics in early stages to prescriptive actuation in management and harvesting, with feedback loops remaining underdeveloped. These cross-stage patterns inform the detailed discussion that follows.

Table 2. Cross-stage comparison of AI applications by analytical dimensions.

Dimension	Tillage	Sowing	Management	Harvesting	Postharvest
Dominant Data Source	Spectral, IoT	Machine vision, RGB	Multispectral, UAV	RGB-D, tactile	Hyperspectral, RFID
Primary Model Type	ML regression, CNN	Object detection, fuzzy-PID	Segmentation, RL	Pose estimation, grasping	Classification, regression
Typical Deployment	Cloud-assisted edge	Real-time edge	Edge with cloud sync	On-board edge	Packing line edge
Operational Objective	Sensing & prediction	Precision control	Diagnosis & prescription	Autonomous manipulation	Grading & traceability

As summarized in Table 2, the four analytical dimensions exhibit clear stage-dependent patterns. Data sources transition from soil-oriented spectral sensing in tillage to crop-focused RGB and multispectral imagery in sowing and management, culminating in RGB-D and hyperspectral sensing for harvesting and postharvest handling. Model types correspondingly evolve from regression and basic CNNs in early stages to more sophisticated object detection, segmentation, and reinforcement learning in later stages where real-time decision complexity increases. Deployment settings reveal a consistent trend toward edge inference for time-sensitive operations, while cloud resources remain essential for training and long-term analytics. Operational objectives demonstrate a progression from passive sensing and prediction toward active control and autonomous manipulation, though closed-loop feedback between stages remains a persistent gap. This systematic cross-stage comparison provides a structured foundation for understanding both the advances and the fragmentation in current AI-enabled agricultural systems.

In addition to the cross-stage comparison presented in Table 2, representative quantitative performance values and ranges of AI methods across selected TSMH-related tasks and deployment scenarios are summarized in Table 3.

Table 3. Representative quantitative performance values and ranges of AI methods across selected TSMH-related tasks and deployment scenarios.

Stage/Scenario	Representative Task	Key Metric(s)	Reported Value/Range	Dataset/Experimental Context	Main Sources of Variability	Example Ref.
Sowing	Seed detection/monitoring	Relative error	<3.0%	Custom vision dataset	Illumination, seed size variation	[20]
Sowing	Seed-metering monitoring	Monitoring accuracy	>95%	Custom bench-top testbed	Operating speed, seed type	[23]
Harvesting	Fruit maturity assessment	Classification accuracy	99.4%	Multimodal tomato images	Sensor modality, fruit variety	[53]
Postharvest	Tomato quality grading	R^2 (MLR model)	0.87	Vis-NIR spectral dataset	Fruit batch, storage conditions	[54]
Edge deployment	Inference speed (YOLOv8 n)	FPS	52–65 (Jetson Orin NX)	Edge-device deployment benchmark	Model quantization level, hardware platform	[63]

Note: Reported values reflect those presented in the cited studies and mainly correspond to controlled, bench-top, or task-specific experimental/deployment conditions. Field performance may be lower due to environmental variability, including illumination changes, dust, vibration, occlusion, sensor configuration, and hardware constraints.

Toward Integrated Workflows: Emerging Examples. Although fully integrated TSMH systems remain rare, several recent studies demonstrate partial cross-stage integration that points toward future possibilities. For example, soil sensing data from autonomous tillage platforms have been successfully reused to generate variable-rate sowing prescriptions, substantially reducing seed input without compromising yield. Similarly, UAV-based weed maps generated during early-season management have been coupled with harvester-mounted quality sensors to predict postharvest grading outcomes before harvest, enabling selective harvesting strategies. Cloud-based platforms such as FarmBeats have demonstrated closed-loop pipelines where irrigation decisions informed by crop water stress indices are automatically translated into actuator commands, with feedback from yield monitors refining the following season's sowing density models. These examples, while still research-grade, illustrate the feasibility and value of cross-stage data interoperability and provide concrete templates for future system-level integration.

Across the full tillage-sowing-management-harvesting workflow, current AI applications are clearly more mature in stage-level optimization than in whole-chain coordination. In practice, perception, prediction, and control have advanced rapidly within individual tasks, but interoperability across machinery, crops, environments, and downstream decisions remains limited. This imbalance helps explain why many studies report high task accuracy under controlled conditions yet still fall short of end-to-end operational intelligence in real agricultural systems. The core bottleneck is not simply algorithmic performance, but architectural fragmentation: upstream sensing outputs are often not formatted, timed, or validated in ways that can directly support downstream decision and execution modules. From a review perspective, this also means that reported metrics

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should be interpreted cautiously; accuracy gains obtained under narrow datasets or fixed field conditions do not automatically translate into robust cross-stage operational value.

Another major challenge lies in data heterogeneity. Agricultural datasets vary widely across crops, regions, seasons, sensors, and field conditions, which weakens model generalization and increases annotation costs. More importantly, heterogeneity in agriculture is not only statistical but operational: changes in machinery settings, planting density, management targets, and weather windows can alter the meaning of the same visual or spectral pattern. In addition, practical deployment is constrained by limited computational resources, environmental noise, and real-time response requirements, especially in edge-computing scenarios involving wind, dust, occlusion, and illumination variability. These factors make it difficult to transfer models from benchmark datasets to reliable, season-long field service. Future comparative studies should therefore report not only final accuracy, but also deployment conditions, annotation cost, inference latency, and robustness to disturbances. Additionally, they should indicate whether the model output can be directly consumed by a downstream controller or management module.

Beyond algorithmic accuracy, practical deployment of sowing-monitoring vision systems is constrained by environmental robustness and computational cost. Lightweight variants such as YOLOv8 n have been shown to reach approximately 52 FPS on an NVIDIA Jetson Orin NX, with INT8 quantization further improving throughput to around 65 FPS, highlighting the speed–resource trade-off in edge deployments [63]. In real field conditions, vision systems are significantly affected by tractor vibrations and dust accumulation on camera lenses, despite achieving very low miss-detection rates in bench-top experiments [64]. Additionally, hardware costs remain a primary constraint for smallholder adoption, emphasizing the need for cost-effective solutions [65]. These deployment realities underscore the urgent need for lightweight model architectures, hardware-software co-design, and sensing suites that balance accuracy, latency, and economic viability.

Future research should therefore focus not only on improving task-specific accuracy, but also on developing multimodal agricultural foundation models, lightweight cloud-edge collaborative architectures, and interoperable full-chain decision systems. Particular attention should be given to four priorities: first, standardized data interfaces that allow upstream sensing results to be reused across sowing, management, harvesting, and traceability modules; second, uncertainty-aware decision pipelines that distinguish confident recommendations from high-risk predictions under field variability; third, closed-loop evaluation frameworks that measure whether perception outputs actually improve subsequent control quality, resource-use efficiency, or harvest outcomes; and fourth, feedback mechanisms that allow results from management and harvest stages to refine upstream models for sowing and field preparation. The long-term value of AI in agriculture will ultimately depend on whether isolated intelligent modules can be transformed into coordinated, adaptive, and scalable production ecosystems that support continuous learning rather than one-off task execution.

Limitations of the Review

This review has several limitations that should be acknowledged. First, the literature selection was constrained to English-language publications, which may omit significant advances reported in regional languages, particularly in leading agricultural research communities in China, Japan, and parts of Europe. Second, due to substantial heterogeneity in evaluation metrics across the reviewed studies—including varying IoU thresholds, inconsistent definitions of "real-time" performance, and diverse dataset characteristics—a formal quantitative meta-analysis was not feasible. Third, the rapid pace of AI development means that very recent model architectures (e.g., Mamba-based vision models, large multimodal foundation models released in late 2025) may be underrepresented in this synthesis, as the search period concluded in December 2025 and the publication pipeline introduces additional latency. Fourth, the review's stage-based TSMH framework, while useful for organizing the vast literature, may inadvertently underemphasize cross-cutting themes such as data privacy, cybersecurity in connected agricultural systems, and socio-economic barriers to technology adoption. Future updates to this review should incorporate standardized benchmark comparisons and expanded linguistic coverage as the field matures.

To provide a holistic view of AI integration across the production-to-postharvest continuum, Figure 5 presents a conceptual framework that synthesizes the core stages, representative AI tasks, data flow layers, and cross-stage feedback mechanisms discussed throughout this review.

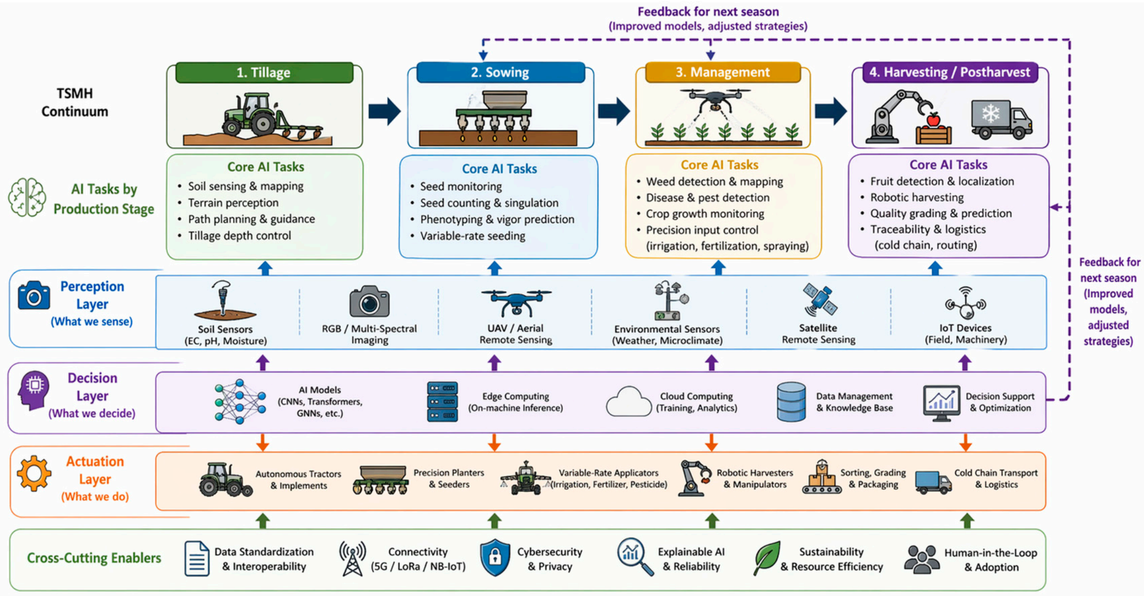

Figure 5. Conceptual framework of AI-enabled smart agriculture across the TSMH continuum.

8. Conclusions and Future Perspectives

This review has systematically examined the application of artificial intelligence across the tillage–sowing–management–harvesting continuum in smart agriculture. The literature confirms that AI is no longer confined to isolated decision-support tools, but is increasingly embedded into core agricultural operations, enabling high-resolution soil sensing, adaptive machinery control, real-time sowing monitoring, crop growth diagnosis, variable-rate input management, robotic harvesting, and non-destructive quality grading. Across these stages, AI has demonstrated measurable improvements in perception accuracy, operational efficiency, and resource-use precision, signaling a broader transition from mechanically assisted farming toward data-driven, intelligence-enabled production systems.

Despite these advances, several structural bottlenecks continue to limit the scalability, reliability, and integration of AI in real-world agricultural settings. Agricultural data remain highly heterogeneous across crops, regions, sensors, and management practices, which undermines model generalization and inflates annotation costs. Many models perform well under controlled conditions but fail in dynamic field environments due to occlusion, illumination variation, dust, and limited onboard computation. Moreover, most existing systems are optimized for isolated tasks, with weak interoperability across stages—soil sensing outputs are rarely reused in sowing decisions, and harvest feedback seldom refines earlier management models. This fragmentation constrains the transition from task-level intelligence to full-chain autonomy.

Addressing these challenges will require a shift in research priorities, from pursuing marginal gains in benchmark accuracy toward enabling system-level robustness, interoperability, and adaptability. Four interconnected directions merit particular attention:

First, multimodal foundation models tailored to agriculture should be developed to improve generalization across crops, environments, and sensing modalities. Such models could leverage self-supervised learning on large-scale, unlabeled agricultural data to reduce annotation dependence and enable few-shot adaptation in under-represented crops or regions.

Second, lightweight and uncertainty-aware architectures are needed to support real-time, trustworthy deployment on edge devices. Model compression, cloud–edge collaboration, and probabilistic inference can help balance computational efficiency with reliable decision-making under field variability.

Third, standardized datasets and evaluation protocols must be established to enable fair cross-study comparison and reproducible validation. Interoperable data formats, benchmark tasks spanning multiple production stages, and metrics that capture not only accuracy but also robustness, latency, and annotation cost will be essential for progress.

Fourth, full-chain decision architectures should be designed to link sensing, diagnosis, control, and feedback across the production workflow. This includes developing digital twins that integrate crop growth models with real-time sensor streams, and reinforcement learning frameworks that optimize sequential decisions—such as sowing windows, irrigation schedules, and harvest timing—under environmental uncertainty.

Ultimately, the value of AI in agriculture will not be determined by isolated task performance, but by its ability to orchestrate coordinated, adaptive, and scalable production ecosystems. Realizing this vision will require sustained collaboration among agronomists, data scientists, engineers, and farmers, moving beyond proof-of-concept demonstrations toward integrated systems that learn, adapt, and improve across seasons.

To provide a clearer trajectory for research and development, these four directions can be further categorized by feasibility and time horizon. Short-term priorities (1–3 years) include lightweight and uncertainty-aware architectures for edge deployment, as well as cloud–edge collaboration frameworks; these directions benefit from mature model compression techniques (e.g., quantization, pruning) and existing IoT infrastructure, with the primary breakthrough required being robust performance under severe field variability. Mid-term priorities (3–5 years) encompass the establishment of standardized agricultural datasets and interoperable evaluation protocols; progress in this area depends on community-driven annotation efforts and consensus on benchmark tasks that span multiple production stages, rather than on fundamental algorithmic advances alone. Long-term priorities (5+ years) involve the development of multimodal agricultural foundation models and full-chain digital twin architectures; these directions require breakthroughs in self-supervised learning on heterogeneous agricultural data and seamless integration of mechanistic crop growth models with real-time sensor streams. By aligning research efforts with this phased roadmap, the transition from isolated task-level intelligence to coordinated, system-level autonomy can be accelerated.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
TSMH	Tillage-Sowing-Management-Harvesting
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
PID	Proportional-Integral-Derivative
IoT	Internet of Things
UAV	Unmanned Aerial Vehicle
RGB	Red, Green, Blue
MDP	Markov Decision Process
IoU	Intersection over Union
CQI	Comprehensive Quality Index
LAI	Leaf Area Index
1D-CNN	One-Dimensional Convolutional Neural Network
PCA	Principal Component Analysis
EMD	Empirical Mode Decomposition

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